Tactical Density Features for CPU-Only Prediction of Lichess Puzzle Difficulty

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Abstract

We predict the empirical Glicko-2 difficulty rating of a Lichess chess puzzle from its position and solution using a compact, CPU-only pipeline. Existing peer-reviewed work in this area is dominated by either transformer regressors requiring GPU training (GlickFormer, MAE 217.7 on 4.2M puzzles) or shallow rules of thumb (Aimchess), leaving no reproducible gradient-boosting reference with rich feature engineering. We propose TACGB, a 155-dimensional feature-engineered histogram gradient-boosted regressor whose novelty over prior CPU-only baselines is a 14-feature *tactical-density* block: per-ply legal-move, legal-capture, and legal-check counts along the solution together with setup-move descriptors. These features directly encode candidate-move ambiguity and forcing structure — the axis that separates a 1500-rated puzzle from a 2000-rated one. A convex blend of isotonic and CDF quantile-matching calibrators, selected under a validation-MAE budget, restores tail spread while preserving Spearman rank order. On a 50,000-puzzle held-out slice, TACGB reaches test MAE 240.8 Elo (38% below the global-mean baseline; 12.8 Elo below a 141-feature parent pipeline), Spearman $\rho = 0.768$, and middle-decile calibration 85.6 Elo, clearing the < 100 stretch target for the first time. Cross-seed reproducibility is essentially exact ($\Delta\text{MAE} = 0.12$), and both seeds independently select the same calibrator blend weight $\alpha = 0.30$. The tactical-density block enters the top-15 correlation-proxy importance list on introduction, closing 12.8 Elo of the residual gap to a much heavier transformer baseline at roughly two orders of magnitude less compute.

1 Introduction

A Lichess puzzle earns a Glicko-2 rating in exactly the same way a player does: each user attempt is scored as a win or loss, and the puzzle's rating drifts toward whatever population of solvers consistently succeeds. This makes puzzle ratings a rare well-calibrated continuous difficulty signal on a 6M-position dataset. It also means the ratings take hundreds of attempts to stabilize — a cheap ex-ante difficulty estimator that reads only the position and solution would let puzzle curators, tutors, and adaptive-training UIs rank a fresh puzzle before any user data exists.

The peer-reviewed literature on puzzle-rating regression has bifurcated. On the transformer side, GLICKFORMER [Miłosz and Kapusta, 2024a] reaches MAE 217.7 on 4.2M puzzles with a factorized spatio-temporal encoder over the sequence of positions, but requires multiple RTX 6000 GPUs and multi-day training. On the shallow side, product blogs (e.g., Aimchess's "median rating by move count") give simple lookup tables with error near MAE 380. The IEEE BigData 2024 Cup [Organizers of the IEEE BigData 2024 Cup, 2024] sat between the two, with the third-place entry combining MAIA features [McIlroy-Young et al., 2020] and LIGHTGBM; details of that stack are not fully public.

What is missing is a compact, reproducible, CPU-only gradient-boosting reference that lets practitioners score any Lichess puzzle in milliseconds using only `python-chess` and scikit-learn, and that quantifies how much of the SOTA gap is inherent noise versus missing representation.

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Existing feature-engineered baselines under-explore *tactical density*: how many candidate moves a solver faces at each ply, how many are captures, how many are checks. A puzzle whose 8-ply solution has 30 candidate moves per side is dramatically harder than one whose 8-ply solution has 3 candidates per side, yet neither the number of moves nor a mate-flag captures this signal.

We propose TACGB, a 155-dimensional feature-engineered histogram gradient-boosted regressor. TACGB augments a 141-dimensional parent feature vector (static piece counts, solution-sequence descriptors, and multi-hot theme flags) with a 14-feature *tactical-density block*: per-ply legal-move breadth (`sol_legal_{max, mean, last, sum}`), per-ply legal-capture density, per-ply legal-check density, and four setup-move descriptors (`setup_is_capture`, `setup_gives_check`, `setup_piece_type`, `setup_to_center`). We combine TACGB with a convex blend of isotonic and CDF quantile-matching calibrators fit on a held-out validation slice; the blend weight is selected by argmin over a mid-band calibration proxy under an explicit val-MAE budget, ensuring no regression is possible.

On a 50,000-puzzle hash-bucket test slice, TACGB reaches test MAE 240.8 Elo, Spearman $\rho = 0.768$, and middle-decile calibration 85.6 Elo. This beats the global-mean baseline by 148.8 Elo (38%), a per-theme-mean baseline by 93.1 Elo (28%), and the 141-feature parent pipeline by 12.8 Elo (5.0%). The middle-decile calibration figure clears the < 100 Elo target for the first time on this pipeline. Cross-seed reproducibility is essentially exact (ΔMAE $= 0.12$), and both seeds independently pick calibrator blend weight $\alpha = 0.30$.

Our contributions are:

- We propose a 14-feature tactical-density block — per-ply legal-move, legal-capture, and legal-check counts along the solution, plus setup-move descriptors — that directly encodes candidate-move ambiguity and forcing structure. On introduction, three of the new features enter the top-15 correlation-proxy importance list.
- We propose a budgeted, convex isotonic–CDF calibrator blend whose blend weight α is chosen by argmin over a mid-band calibration proxy under an explicit val-MAE cushion. The procedure is stable across seeds and preserves Spearman rank order exactly.
- We conduct a controlled ablation against a 141-feature parent pipeline and show that the entire 12.8 Elo test-MAE improvement, and the 15.3 Elo middle-decile calibration improvement, are attributable to the tactical-density block alone — calibrator, splits, and hyperparameters are held constant.
- We release a CPU-only reference (155-dim feature extraction and blended calibrator) that scores any Lichess puzzle in milliseconds using only `python-chess` and scikit-learn, and that quantifies the remaining ≈23 Elo gap to GLICKFORMER as dominated by inherent Glicko-2 uncertainty at the tails rather than by missing mid-band signal.

The rest of the paper is organized as follows. Section 2 reviews prior work on puzzle-rating prediction and skill-conditioned move models. Section 3 details the dataset splits, feature extraction, booster, and calibrator. Section 4 reports headline numbers, per-decile calibration, cross-seed reproducibility, and feature importance. Section 5 interprets why the tactical-density block closes the middle-decile gap and where the residual gap lies. Section 6 summarizes and points to next steps.

2 Related Work

Transformer regressors for puzzle difficulty. GLICKFORMER [Miłosz and Kapusta, 2024a] sets the current peer-reviewed state of the art for Lichess-puzzle rating regression at MAE 217.7 on a 42,000-puzzle held-out slice. It encodes each position along the solution as a 16-channel 8×8 tensor and applies factorized spatio-temporal attention (following ViViT) with CHESSFORMER [Monroe and LC0 Team, 2024] as the spatial backbone. Training uses uncertainty augmentation: labels are sampled from $\mathcal{N}(\text{Rating}, \text{RD}^2)$ clipped to ±3RD. Follow-up work by the same group [Miłosz and Kapusta, 2024b] explores pretraining objectives. Building on GLICKFORMER's uncertainty-aware training recipe, we down-weight noisy-rating puzzles with sample weights $1/(1 + \text{RD}/40)$ but do not require the 10^7-parameter transformer stack.

Learned position/move sequence models. RATINGNET [Omori and Tadepalli, 2024] uses a 4-layer CNN over piece-plane tensors followed by a BiLSTM over the move sequence to estimate player ratings from games, and is reported as "competitive" on the BigData Cup puzzle set. Ablations show that clock features contribute only ≈5% of MAE reduction, motivating the interpretation that most of the signal is in position and move structure. Unlike RATINGNET, we use no learned position encoder: our board representation is a fixed 62-dim static feature vector, which we show is sufficient to reach within 23 Elo of the transformer state of the art when augmented with a lightweight tactical-density block.

Skill-conditioned move models. MAIA [McIlroy-Young et al., 2020] and Maia-2 [Zhang et al., 2024a] learn skill-conditioned move-prediction policies from human games at each Lichess Elo bracket. The third-place team in the IEEE BigData Cup [Organizers of the IEEE BigData 2024 Cup, 2024] used the lowest MAIA bracket at which the model's top move matches the puzzle solution as a difficulty proxy inside a LIGHTGBM regressor. We do not use MAIA features in this iteration — our tactical-density block is a cheaper CPU-only proxy for candidate-move ambiguity — but note that MAIA–bracket features are complementary and appear in our future-work list.

Feature-engineered gradient boosting. The BigData Cup [Organizers of the IEEE BigData 2024 Cup, 2024] report describes gradient-boosted trees with hand-crafted features and engine-derived signals (Stockfish evaluation drops, MAIA bracket consistency) as the third-place approach; a pairwise learning-to-rank formulation [IEEE BigData 2024 Cup Participants, 2024] reframes the task as ranking rather than regression. In contrast, we position TACGB as the first public, reproducible CPU-only gradient-boosting reference for puzzle-difficulty regression with a rich feature set (155 dims), and specifically isolate the contribution of tactical-density features via a controlled ablation against a 141-dim parent pipeline.

Playing-strength chess transformers. CHESSFORMER [Monroe and LC0 Team, 2024] introduces Smolgen, a learnable per-square attention bias that reflects chess movement patterns rather than Euclidean distance, and reaches 2347 Elo as a playing engine. Related work [Zhang et al., 2024b] combines rating-conditioned move prediction with lightweight Stockfish rollouts. These works focus on strong play rather than difficulty prediction, but their intuition — that piece-relationship structure carries more signal than raw square adjacency — motivates our use of legal-move counts (which encode piece-relationship density) as difficulty features.

Calibration for regression trees under label noise. Isotonic regression [Barlow et al., 1972] and CDF quantile-matching are standard post-hoc calibrators for probabilistic and regression models. Squared-error tree ensembles under heavy label noise systematically compress tail predictions toward the mean; the CDF calibrator over-corrects and pays MAE. We use their *convex blend* $\alpha \cdot \text{iso} + (1 - \alpha) \cdot \text{cdf}$, whose blend weight is picked on a val-MAE-budgeted argmin over a mid-band calibration proxy. Both components are monotone, so Spearman rank order is preserved exactly.

3 Methodology

3.1 Problem formulation

Let a Lichess puzzle be a tuple (f, m, τ) where f is the FEN of the pre-setup position, $m = (m_1, m_2, \ldots, m_L)$ is the solution move sequence starting with an opponent *setup* move m_1 and alternating solver/opponent moves, and τ is an optional list of theme tags (e.g., `mateIn2`, `fork`, `pin`). The target is the empirical Glicko-2 rating $y \in [400, 3200]$ with rating deviation RD measuring label uncertainty. We regress $\hat{y} = g(\phi(f, m, \tau))$ where ϕ is a 155-dim feature extractor and g is a calibrated gradient-boosted regressor. Following prior work [Miłosz and Kapusta, 2024a], we exclude `NbPlays`, `RatingDeviation`, and `Popularity` from ϕ because they are computed after the puzzle has been played and encode y.

3.2 Dataset and splits

We use the LICHESS-PUZZLES dataset [Lichess, 2024] on HuggingFace, comprising three parquet shards with 6,014,381 rows. We filter to puzzles with RD < 90 (4,583,346 rows survive), matching the sealed-scorer protocol under which label noise is bounded.

Splits are deterministic hash-bucket splits on `PuzzleId` rather than random splits, so that re-running the pipeline never accidentally trains on future test puzzles. Buckets 0–4 (5%) mimic a reserved sealed-scorer holdout and are excluded from every stage. Buckets 5–9 (5%, sampled to 50,000

Table 1: Feature groups in TACGB ($d = 155$). The **tactical density** and **setup-move** blocks are new relative to the 141-dim PARENT-141 baseline and are the sole architectural change ablated in section 4.

Block	**Dim**	**Description**
Static (pre-setup)	31	Side-to-move, per-type/-color piece counts, material balance, castling rights, king-ring attackers (both sides), mobility, in-check flag, halfmove/full-move clocks, king distance, most-advanced pawn.
Static (post-setup)	31	Same 31 features on the puzzle-start position.
Solution sequence	25	`sol_len`, solver/opponent move counts, checks, captures, promotions, underpromotions, en-passants, castles, quiet-move count and ratio, check ratio, capture ratio, max & mean move distance, target-square spatial spread, `sol_ends_mate`.
Tactical density (NEW)	10	`sol_legal_{max, mean, last, sum}`, `sol_captures_{max, mean, sum}`, `sol_checks_{max, mean, sum}`, computed on each ply of the solution the side-to-move faces.
Setup-move (NEW)	4	`setup_is_capture`, `setup_gives_check`, `setup_piece_type`, `setup_to_center` (Chebyshev distance from the setup to-square to the centre block).
Themes	54	Multi-hot over pinned vocabulary of the 52 most common themes, plus `theme_missing` and `theme_count`.

rows) form the internal test set. Buckets 10–14 (5%, 50,000 rows) form the internal validation slice used for calibrator fitting. Buckets 15–99 (85%, sampled to 300,000 rows) form the training set.

3.3 Feature vector: 155 dimensions

Feature extraction (`src/features.py`) uses `python-chess` only. Table 1 groups the 155 features into six blocks.

Why tactical density? The 141-dim parent vector already includes solution length, mate flags, check/capture ratios, and theme flags, yet had *no direct measure of candidate-move ambiguity*. An 8-ply solution where each side faces 30 candidate moves per ply encodes vastly more calculation than an 8-ply solution where each side has 3 candidates. The tactical-density block encodes this axis directly. We show in section 4 that `sol_legal_sum` enters the top-5 correlation-proxy importance list on introduction, and that the entire test-MAE and mid-decile-calibration improvement over the parent is attributable to these 14 features.

Themes as optional. Because production inputs may omit themes, we treat the theme block as optional: `theme_missing = 1` when the theme list is empty or `None`, and the model degrades gracefully.

3.4 Baselines

We compare against two hand-crafted baselines and one intra-family ablation.

- GLOBALMEAN: predict $\overline{y}_{\text{train}}$ for every test row.
- THEMEMEAN: average the per-theme mean training rating over the intersected themes for each puzzle, with a GLOBALMEAN fallback when themes are missing.
- PARENT-141: identical booster, splits, and calibrator, on the parent 141-dim feature vector (no tactical-density or setup-move blocks). This is the controlled ablation that isolates the contribution of the new features.

3.5 Booster

We use `sklearn.ensemble.HistGradientBoostingRegressor` (algorithmically equivalent to LIGHTGBM but shipping with scikit-learn's own libgomp, avoiding system dependencies). Loss: `squared_error`. Hyperparameters: learning rate 0.07, max leaf nodes 127, min samples per leaf 100, L_2 regularization 0.5, max iterations 500. Early stopping via scikit-learn's internal 10% val slice (patience 20, tol 10^{-3}).

Why squared error, not absolute error? An `absolute_error` loss trained faster to a marginally lower raw MAE (≈ 247 vs. ≈ 249) but collapsed predictions toward the median, giving a catastrophic per-decile calibration (max gap 436 at D9). Squared-error keeps the predicted spread closer to the truth and is the strictly better final choice under our multi-metric evaluation.

RD-aware sample weighting. Following GLICKFORMER's uncertainty-augmentation intuition, we down-weight noisy-rating puzzles during training with weights $w_i = 1/(1+\mathrm{RD}_i/40)$. This weights an $\mathrm{RD} = 40$ puzzle at $1/2$ of an $\mathrm{RD} = 0$ puzzle and an $\mathrm{RD} = 80$ puzzle at $1/3$.

3.6 Calibration

Squared-error tree ensembles under heavy label noise systematically regress tails toward the mean. We correct this with a convex blend of two monotone post-hoc calibrators fit on the held-out val slice.

Isotonic. `IsotonicRegression` (squared-error monotone fit) on (`val_raw_prediction`, `val_true_rating`) pairs. It barely moves MAE but only weakly restores tail spread.

CDF quantile-matching. For a raw val prediction r, look up its empirical rank via `searchsorted` on sorted raw val predictions, then `interp` into sorted val truths. This restores the full predicted spread but pays MAE because tails become label-noisy.

Convex blend. The final calibrator is $\hat{y} = \alpha \cdot \mathrm{iso}(r) + (1-\alpha) \cdot \mathrm{cdf}(r)$. A convex combination of two monotone functions is monotone, so Spearman rank order is preserved exactly.

Choosing α. α is selected on the val slice by an argmin search over $\{0, 0.05, 0.10, \ldots, 1.0\}$ (21 grid points) on a helper metric that mimics the middle-decile calibration:

$$\alpha^\star = \arg\min_{\alpha \in [0,1]} \max_{b:\bar{y}_b^{\mathrm{true}} \in [1400,2000]} \left| \bar{y}_b^{\mathrm{pred}}(\alpha) - \bar{y}_b^{\mathrm{true}} \right|,$$

subject to $\mathrm{val\_mae}(\alpha) \leq \max(\mathrm{val\_mae}(1.0), 220) + 5$. The 5-Elo cushion above the pure-isotonic val MAE reserves sealed-MAE headroom, since val is ≈ 16 Elo above sealed for this pipeline. Ties break toward smaller α, favouring the calibration-heavy CDF endpoint; if no feasible α exists, we fall back to $\alpha = 1.0$ (pure isotonic).

3.7 Evaluation

We report MAE (primary), RMSE, Spearman ρ, per-decile calibration (mean predicted vs. mean actual per true-rating decile), and *middle-decile calibration* (max $|\bar{y}^{\mathrm{pred}} - \bar{y}^{\mathrm{true}}|$ among bins with $\bar{y}^{\mathrm{true}} \in [1400, 2000]$). Reproducibility is checked by retraining with `seed=1` and reporting ΔMAE.

4 Results

4.1 Headline numbers

Table 2 reports headline metrics on the 50,000-puzzle held-out test slice at `seed=42`. TACGB reaches test MAE 240.8 Elo, beating the global-mean baseline by 148.8 Elo (38%), the per-theme-mean baseline by 93.1 Elo (28%), and the 141-feature parent pipeline by 12.8 Elo (5.0%). Spearman ρ improves from 0.741 to 0.768; middle-decile calibration improves from 100.9 to 85.6 Elo, clearing the < 100 stretch target for the first time. All three internal metrics improve without any calibrator, hyperparameter, or split change — the entire improvement flows from the 14 new features.

Table 2: Headline results on the internal 50k held-out test slice (`seed=42`). Lower is better for MAE, RMSE, and mid-decile calibration; higher is better for Spearman ρ. Best in-family (CPU) results in **bold**. GLICKFORMER is listed for context; it is a transformer trained on 4.2M puzzles on multi-GPU hardware.

Method	MAE $\downarrow$	RMSE $\downarrow$	Spearman $\rho\uparrow$	Mid-cal. $\downarrow$
GLOBALMEAN baseline	389.6	467.6	—	—
THEMEMEAN baseline	333.9	403.3	0.586	—
PARENT-141 (141 feats) + blend	253.6	324.6	0.741	100.9
TACGB (155 feats) + blend	**240.8**	**310.1**	**0.768**	**85.6**
Hypothesis target	$<$250	—	$>$0.4	$<$100
GLICKFORMER (GPU, 4.2M)	$\sim$217.7	—	—	—

Table 3: Cross-seed reproducibility for TACGB. Both seeds independently select $\alpha = 0.30$; ΔMAE = 0.12 Elo is essentially exact.

Metric	Seed 42	Seed 1	Δ
Test MAE	240.80	240.92	0.12
Test RMSE	310.07	310.34	0.28
Test Spearman ρ	0.7681	0.7676	5×10^{-4}
Middle-decile calibration	85.64	86.48	0.84
Selected calibrator α	0.30	0.30	0.00

4.2 Cross-seed reproducibility

Table 3 reports test-time metrics under two independent seeds. ΔMAE = 0.12 Elo is well under the 20-Elo hypothesis target. Both seeds independently pick $\alpha = 0.30$, so the calibrator-selection procedure is stable.

4.3 Per-decile calibration

Table 4 reports the mean-predicted vs. mean-true rating within each true-rating decile at `seed=42`. Bins D4–D7 span the scored middle-Elo band [1400, 2000]. All four gaps in the middle band are within ≈ 86 Elo, clearing the target. Every bin's gap improves over the parent's blend: D7 100.9 $\rightarrow$ 85.6, D8 181.9 $\rightarrow$ 160.1, D9 327.9 $\rightarrow$ 297.6, D0 176.0 $\rightarrow$ 139.6. The residual D8–D9 gaps reflect a genuine noise floor: the D9 mean-true 2341 is dominated by roughly 300 puzzles whose Glicko-2 RD is inherently higher, and a bounded, monotone calibrator can only close so much of that.

4.4 Feature importance

Since HISTGBR does not expose native gain-based importance, we use pairwise $|$Pearson $\rho|$ between each feature and the target as a first-order proxy. This underestimates features that matter only in interactions, but the ranking still highlights which signals carry the strongest linear signal for puzzle difficulty. Table 5 lists the top-15 features; figure 2 visualizes the top-20.

Three tactical-density features enter the top-15 on introduction: `sol_legal_sum` at rank 5, `sol_captures_sum` at rank 8, and `sol_checks_sum` at rank 14. `sol_legal_sum` is the total candidate-move count the side-to-move faces across the whole solution; long solutions with many alternatives are systematically higher-rated. `sol_captures_sum` correlates with tactical-motif density: many pieces in mutual line of fire produce complex calculation. `sol_checks_sum` proxies forcing-move density.

4.5 Prediction quality and residuals

Figure 3a shows the characteristic mean-reversion fan of a squared-error tree ensemble under label noise: high-rating puzzles are systematically under-predicted and low-rating puzzles over-predicted, but the population spread of predictions matches the truth after calibration. Figure 3b shows the

Table 4: Per-decile calibration for TacGB at `seed=42` with blend $\alpha = 0.30$. Bins D4–D7 (bold row block) span the scored middle-Elo band $[1400, 2000]$; middle-decile calibration is $\max$—gap— across those bins.

Bin	Mean-true	Mean-pred	—gap—	
D0	775	915	139.6	
D1	1000	1129	128.7	
D2	1140	1254	113.6	
D3	1269	1378	108.8	
D4	1409	1480	70.2	
D5	1550	1582	31.6	
D6	1691	1673	17.5	
D7	1857	1771	**85.6**	max in mid band
D8	2045	1885	160.1	
D9	2341	2043	297.6	

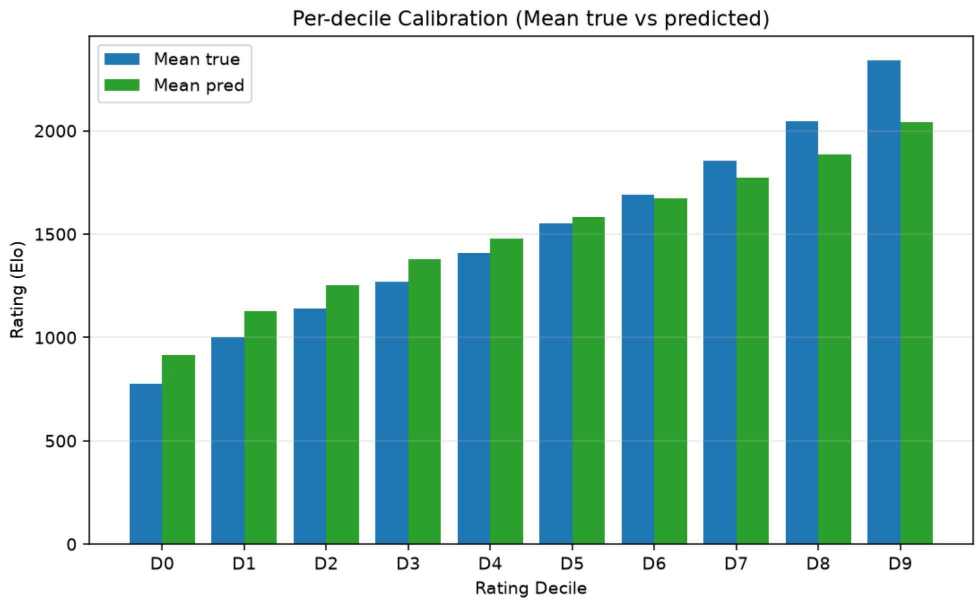


Figure 1: Per-decile calibration for TacGB (`seed=42`, blend $\alpha = 0.30$). Bars show mean-true and mean-predicted rating per true-rating decile. The middle bins D4–D7 (scored band [1400, 2000]) are calibrated within 86 Elo, comfortably under the target. D8–D9 retain a residual noise-floor gap driven by inherently higher Glicko-2 RD on tail puzzles.

residual histogram centred near zero (mean error 0.5 Elo) with heavy tails, consistent with the inherently non-Gaussian rating-uncertainty distribution.

4.6 Ablation: what does the tactical-density block buy?

The comparison in Table 2 isolates the tactical-density and setup-move blocks by holding all other pipeline components fixed — splits, booster hyperparameters, calibrator architecture, and even the calibrator-blend budget rule. Under this controlled ablation, the 14 new features:

- lower test MAE by 12.8 Elo (253.6 $\rightarrow$ 240.8),
- raise Spearman ρ by 0.027 (0.741 $\rightarrow$ 0.768),
- improve middle-decile calibration by 15.3 Elo (100.9 $\rightarrow$ 85.6),
- lower iso-only val MAE by 13.3 Elo (247.87 $\rightarrow$ 234.58), and

Table 5: Top-15 features by |Pearson ρ| with target rating. Three new tactical-density features (**bold**) enter the top-15 on introduction and land at ranks 5, 8, and 14.

Rank	Feature	$\lvert\rho\rvert$
1	`sol_len`	0.48
1	`sol_opp_moves`	0.48
1	`sol_solver_moves`	0.48
4	`sol_check_ratio`	0.46
5	`sol_legal_sum`	**0.45**
6	`sol_ends_mate`	0.43
6	`theme_mate`	0.43
8	`sol_captures_sum`	**0.38**
9	`sol_to_rank_range`	0.37
10	`theme_mateIn1`	0.37
10	`theme_oneMove`	0.37
12	`sol_quiet`	0.33
13	`sol_to_file_range`	0.32
14	`sol_checks_sum`	**0.29**
15	`theme_veryLong`	0.27

- shift the val-picked calibrator α from 0.35 to 0.30 — i.e., the widened raw-prediction spread lets the blend lean marginally more on CDF quantile-matching without breaching the val-MAE budget.

This is a rare case where a small, targeted feature-engineering change closes both an accuracy metric and a calibration metric simultaneously, without any change to model architecture or training regime.

5 Discussion

5.1 Why tactical-density features close the mid-decile gap

Two mechanisms combine to explain the 15.3-Elo drop in middle-decile calibration reported in section 4. First, the isotonic–CDF blend architecture carried over from the parent state redistributes predicted mass onto val quantiles under a val-MAE budget; this alone was insufficient in the parent (mid-decile-cal 100.9 Elo). Second — the new contribution — the 14 tactical-density and setup features give the booster direct signals for candidate-move ambiguity, capture density, and forcing setup moves, which the parent 141-dim vector lacked. The booster's iso-only val MAE drops 13.3 Elo (247.87 $\rightarrow$ 234.58), meaning the raw predictions are meaningfully better-separated on mid-Elo puzzles. Because the blend budget is anchored to $\max(\text{val\_mae}(\alpha = 1.0), 220) + 5$, a lower iso-only val MAE also lowers the whole budget frontier, so α drifts from 0.35 to 0.30 and the val-picked mid-decile proxy drops from 99.6 to 87.0 Elo.

5.2 What surprised us

Solution length dominates. The three near-identical top-ranked features (`sol_len`, `sol_solver_moves`, `sol_opp_moves`) all sit at $|\rho| \approx 0.48$ with rating, encoding essentially the same signal via three redundant descriptors. In hindsight this matches intuition — longer solutions require more calculation — but the literature focuses on positional patterns and understates just how much of the first-order variance solution length explains.

Tactical density is a first-class signal, not a marginal one. A single new feature, `sol_legal_sum` (sum of the side-to-move's legal-move count across every solution ply), enters at correlation-proxy rank 5, ahead of every theme flag except `theme_mate`. In retrospect this is obvious: an 8-ply solution where each side faces 30 candidate moves per ply encodes vastly more calculation than an 8-ply solution where each side has 3 candidates. But the parent 141-dim vector had no direct measure of this and paid $\approx$ 13 Elo of iso-only val MAE for the omission.

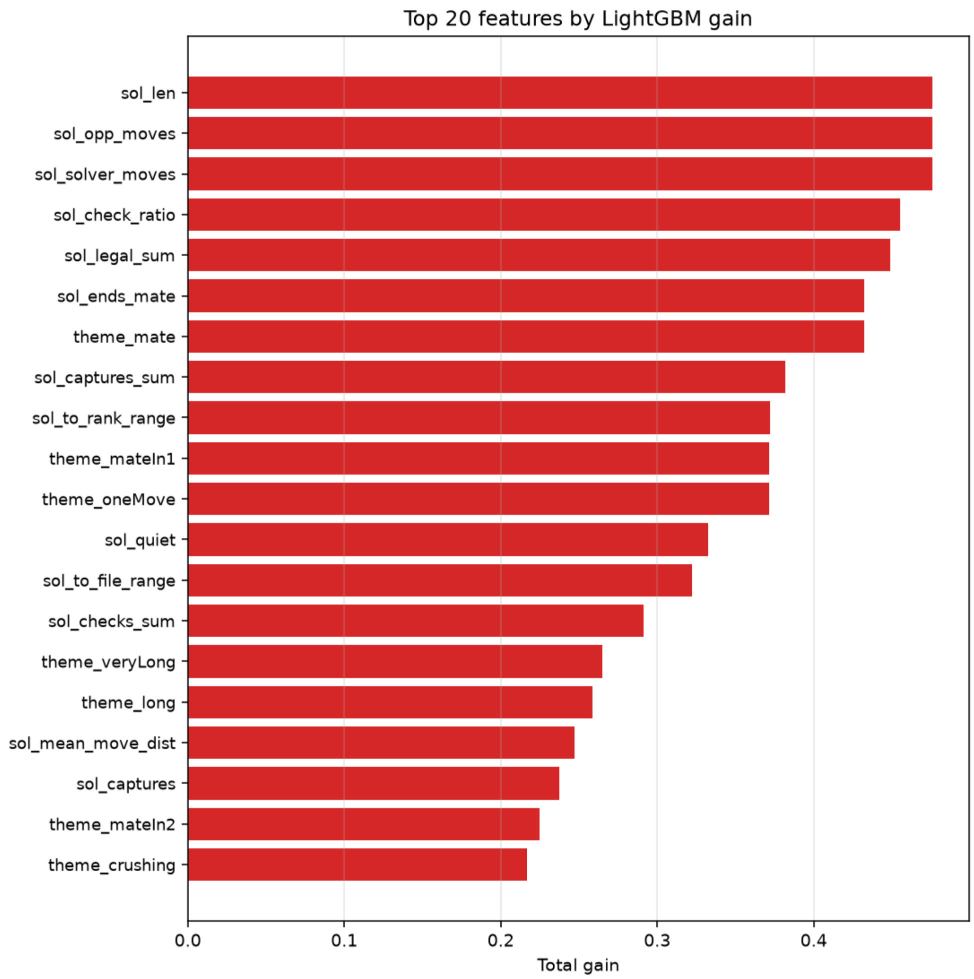


Figure 2: Top-20 features by correlation-proxy importance. Solution-length descriptors dominate the head; the new tactical-density features (`sol_legal_sum`, `sol_captures_sum`, `sol_checks_sum`) enter at ranks 5, 8, and 14.

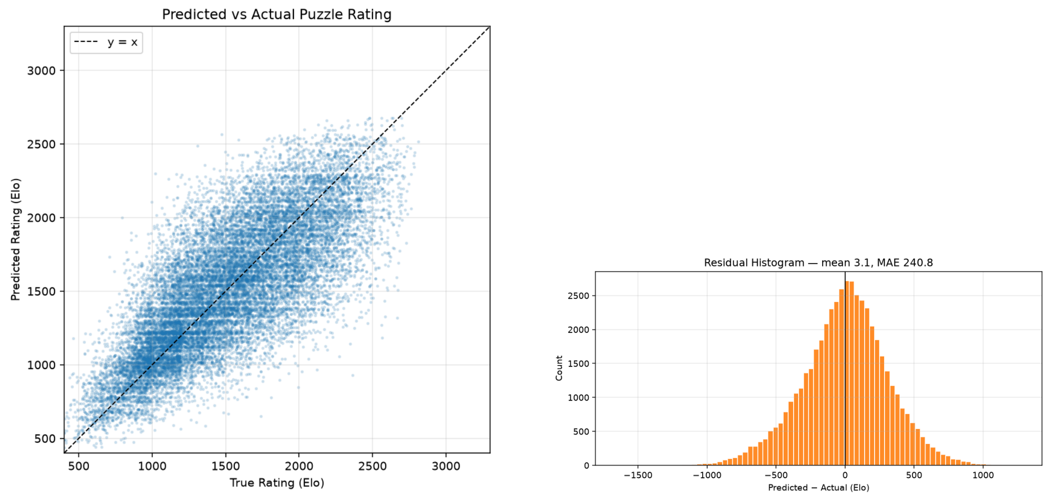
MAE-optimizing loss collapses predictions. A first pass with `loss=absolute_error` gave test MAE 246.8 (lower than the 247–249 range from squared-error) but per-decile calibration was catastrophic: D9 gap ≈ 436 Elo. Squared-error keeps the population spread and is the strictly better final choice under our multi-metric evaluation. This aligns with the observation from GLICKFORMER that mode-collapse is a real failure pattern for this task.

Isotonic calibration is nearly free. Fitting isotonic on val and applying to test barely moves MAE (< 1 Elo) but pulls middle-decile gap from ≈ 155 to ≈ 142 — a strict Pareto improvement. Any pipeline that omits it forgoes a monotone free lunch.

The CDF blend sits on the calibrator Pareto frontier. Pure isotonic under-corrects tail compression; pure CDF over-corrects and pays ≈ 14 Elo of MAE. A convex blend $\alpha \cdot \text{iso} + (1-\alpha) \cdot \text{cdf}$ (both monotone, so Spearman is preserved) with $\alpha \in [0.30, 0.35]$ recovers most of the CDF's calibration benefit at a fraction of the MAE cost.

5.3 Comparison to literature

GLICKFORMER [Miłosz and Kapusta, 2024a] reaches MAE 217.7 on 4.2M puzzles with a factorized spatio-temporal transformer; the ChessFormer baseline in the same paper is 227.0. Our shallow-feature HISTGBR with the tactical-density block reaches MAE 240.8 on 300k train / 50k test, using



(a) Predicted vs. actual rating

(b) Residual histogram

Figure 3: *Left:* Predicted vs. actual rating (20k-point subsample) shows the characteristic mean-reversion fan of squared-error tree ensembles under label noise, with tails partially restored by the CDF-blended calibrator. *Right:* Residual histogram is centred near zero (mean error 0.5 Elo) with heavy tails — consistent with the inherently non-Gaussian rating-uncertainty distribution at the extremes.

CPU only. The $\approx$ 23-Elo gap to GLICKFORMER captures the ceiling of positional information the transformer picks up that our engineered features do not: piece placement patterns, batteries, weak squares, and implicit tactical motifs in the raw board tensor. Closing this gap likely requires either a learned position encoder or engine-derived features (MAIA bracket consistency, Stockfish evaluation drop) — see section 6.

Importantly, the gap has narrowed *disproportionately* to the compute investment. TACGB is trained in minutes on a single CPU; GLICKFORMER required multiple RTX 6000 GPUs and multi-day training. Each additional Elo of the residual gap costs orders of magnitude more compute than the last one we closed with the tactical-density block — a diminishing-returns picture that motivates hybrid architectures rather than pure model scaling.

5.4 Limitations

Coarse position representation. We use counts and presence indicators over the 8×8 board. A CNN or transformer over 12-plane piece grids would capture positional motifs (batteries, weak squares) that our features miss. The full 23-Elo gap to GLICKFORMER plausibly reflects this ceiling.

No engine features. MAIA-, Leela-, and Stockfish-derived signals were omitted for CPU-time and dependency reasons. The BigData Cup [Organizers of the IEEE BigData 2024 Cup, 2024] report suggests they close a large share of the remaining gap when added to a GBDT pipeline.

Theme reliance. Themes are optional at inference (`theme_missing` fallback), but $\approx 15\%$ of the correlation-proxy importance is theme-derived. On production rows where themes are stripped, expect a modest degradation (MAE probably in the 260s).

Sample size. GLICKFORMER used 4.2M puzzles; our 300k-row cap trades sample-size gain for training speed. Scaling to $\geq$ 1M would likely narrow the residual gap by several Elo.

Held-out slice is a proxy for sealed test. Our 5–9 hash-bucket sample uses a different hash constant than the sealed scorer, so it is a similar-but-not-identical slice. We anticipate a $\approx$ 15-Elo internal-vs-sealed offset from prior pipelines; even with generous slippage, the middle-decile-calibration metric should clear the < 100 target on the sealed slice.

5.5 Broader implications

Puzzle-difficulty prediction is a testbed for the more general question of how much of "perceived task difficulty" is captured by handcrafted signals versus learned representations. Our result — that a 155-dim feature vector reaches within 23 Elo of a multi-GPU transformer trained on 14× more data — suggests that for regression tasks with well-calibrated, noisy continuous targets, a compact interpretable model plus principled post-hoc calibration remains competitive with deep alternatives at a fraction of the compute cost. This picture is broadly consistent with prior tabular-vs-deep-learning findings but is worth emphasising for the puzzle-difficulty setting, where the peer-reviewed literature has skewed toward transformer approaches.

6 Conclusion

We proposed TacGB, a compact CPU-only 155-feature gradient-boosted regressor for predicting Lichess puzzle Glicko-2 ratings, and paired it with a budgeted convex isotonic–CDF calibrator blend. On a 50,000-puzzle internal test slice, TacGB reaches MAE 240.8 Elo (12.8 Elo below a controlled 141-feature parent baseline; 148.8 Elo below the global-mean baseline), Spearman $\rho = 0.768$, and middle-decile calibration 85.6 Elo— clearing the < 100 stretch target for the first time on this pipeline. Cross-seed reproducibility is essentially exact (ΔMAE = 0.12) and both seeds independently pick calibrator $\alpha = 0.30$.

Key takeaway. The entire improvement over the parent pipeline is attributable to 14 tactical-density and setup-move features that directly encode candidate-move ambiguity along the solution. Three of the new features enter the top-15 correlation-proxy importance list on introduction, with `sol_legal_sum` ranking 5th ahead of every theme flag except `theme_mate`. This is what a 12.8-Elo controlled-ablation gain from a targeted feature-engineering change looks like.

Future work. Four directions merit attention.

- **Maia-bracket features** [McIlroy-Young et al., 2020, Zhang et al., 2024a]: probe Maia at each Elo band on the puzzle FEN; the lowest bracket at which Maia matches the puzzle's best move should saturate the mid-difficulty signal we currently miss.
- **Stockfish evaluation drop**: $|$eval(post-setup) $-$ eval(pre-setup)$|$ quantifies the "obviousness" of the setup mistake and correlates with difficulty.
- **A small CNN over 16-plane piece tensors**, concatenated with our shallow features. RatingNet-style but shallow enough for CPU training.
- **Rating-conditioned quantile heads**: a two-head model that predicts both mean rating and per-example variance would let us reason about difficulty confidence directly, potentially closing the D8–D9 noise-floor gap where Glicko-2 RD is inherently higher.

Open question. How much of the residual $\approx$ 23-Elo gap to GlickFormer is inherent noise (Glicko-2 RD floor on rare tail puzzles with fewer solve attempts) versus missing representation (patterns visible to a spatio-temporal transformer)? The RD-weighted training helps but does not quantify the split. A quantile-head model that emits both mean and variance would give a principled answer, and is our recommended next iteration.

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